

DAWOOD UNIVERSITY OF ENGINEERING & TECHNOLOGY
FINAL EXAMINATION OF 1ST SEMESTER, 1ST YEAR
FALL SEMESTER 2025 OF BS ARTIFICIAL INTELLIGENCE
SECTIONS (A1, A2 & A3)

APPLIED PHYSICS: (THEORY)

MAX. MARKS 50

DATED: 14-01-2026

TIME ALLOWED 03 HOURS

Student Id Number: _____

Student Name: _____

Instructions:

1. This paper contains 05 questions.
2. Use of a Calculator is allowed. Use of a mobile phone as a calculator is not allowed.
3. Cheating of any type will disqualify the candidate.
4. Do not detach the sheets. (Paper will be cancelled if the sheets are detached)
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9. The Answer Script must be returned to the invigilator before leaving the exam hall.

NOTE: - ATTEMPT ALL THE FOLLOWING QUESTIONS AND EXPLAIN EACH STEP FOLLOWED

Q.NO. 1 (A) Define briefly the following with their diagrams.

(CLO1, PLO1, 6 MARKS)

- (i) Energy bands in solids.
- (ii) Construction and working of a diode.
- (iii) EMF source and voltage source.

(B) Show that the electric flux due to n charges enclosed in a surface is $\frac{q}{\epsilon_0}$. (CLO2, PLO2, 4 MARKS)

Q.NO. 2 (A) State Gauss's law and, by using it, find the electric field due to an infinite positively charged line.

Explain why the same Gaussian surface (cylindrical) cannot be used to find the exact electric field of a positively charged line at a general point. (CLO1, PLO1, 5 MARKS)

(B) An electrostatic force of 1.5×10^{-11} N is acting on a positive charge q when placed at 10 cm from an infinite positively charged line with linear charge density 30.0 nC/m. find the magnitude of charge q . (CLO3, PLO3, 5 MARKS)

Q.NO. 3 (A) A parallel plate capacitor is connected to a battery, maintaining a constant potential difference V . If the separation between the plates is doubled while the capacitor remains connected, explain why the magnitude of the electric field between the plates does not change, despite the increased separation. Also, explain the changes that occur to charge stored, capacitance, and voltage when a dielectric material is placed between the plates? (CLO2, PLO2, 5 MARKS)

(B) Two plates with the same area of 10 cm² are separated and fixed at a 5mm separation in air to make a parallel plate capacitor. Find its capacitance, potential difference developed between the plates, and electric field between the plates when a 100 C charge is stored in it. (CLO3, PLO3, 5 MARKS)

Q.NO. 4 (A) Explain briefly the derivation of the relation for potential difference due to an infinite positively charged line, at a distance ' r ' from its centre? (CLO2, PLO2, 5 MARKS)

(B) The potential difference of 10 mV is measured at a distance of 5 cm from the infinite positively charged line, when the reference point is taken at 500 cm from the line. Find the linear charge density of the line and the potential difference at 50 cm, assuming all parameters remain the same. (CLO3, PLO3, 5 MARKS)

Q.NO. 5 (A) Explain the derivation of the relation for finding the magnetic field due to a finite straight current-carrying conductor at point ' p ', from its centre? Hint: use Biot-Savart's law. (CLO2, PLO2, 5 MARKS)

(B) A conductor of length 20 cm is placed vertically, and a current 2 A is allowed to flow vertically upward from it. Find the magnitude and direction of the magnetic field at 5 cm from its centre. (CLO3, PLO3, 5 MARKS)

$$\mu_0 = 4\pi \times 10^{-7} \text{ Tm/A}, k = 9 \times 10^9 \text{ Nm}^2/\text{C}^2, \epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2$$

THE END



CALCULUS & ANALYTICAL GEOMETRY (THEORY)

DATED 12-January-2026

TIME ALLOWED: 3 Hours

MAX. MARKS 50

Student Name: _____

Student Id Number: _____

Instructions:

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NOTE: ATTEMPT ALL THE FOLLOWING QUESTIONS.

Question # 01:

3

a. Interpret whether the given limits are an indeterminate form?

i- $\lim_{x \rightarrow 0} \left(\frac{1}{x} - \cot x \right)$

ii- $\lim_{x \rightarrow 0} \frac{\sin x}{x}$ 1

iii- $\lim_{x \rightarrow \infty} 1^x$

iv- $\lim_{x \rightarrow \pi} \left(\frac{\tan x}{\tan 3x} \right)$

(CLO-1, PLO-1, C2, 4 Marks)

b. Approximate any two limits given in part (a) using L Hospital rule if applicable.

(CLO-1, PLO-1, C2, 6 Marks)

Question # 02:

a. Derive the value of c using the Mean Value Theorem for the function $f(x) = \sqrt{x-2}$ on [2,4].

(CLO-2, PLO-2, C3, 5 Marks)

b. Apply the rules of integration to evaluate the following:

(CLO-3, PLO-2, C3, 5 Marks)

i. $\int e^{-x} dx$

ii $\int_0^2 \frac{x+1}{x^2+2x} dx$

Question # 03:

a. Solve the Maclaurin series of $f(x) = x^4 - 8x^2$

(CLO-2, PLO-2, C3, 5 Marks)

b. Determine the interval on which the function given in part (a) is increasing or decreasing.

(CLO-2, PLO-2, C3, 5 Marks)

Question # 04:

a. In an artificial intelligence system, the activation strength of a neuron in a hybrid neural network depends on two variables:

- x: the input feature intensity
- y: the phase parameter controlling oscillatory behavior in the network

The activation function is modeled as: $f(x, y) = e^x \cos y$

Discover the partial derivative how the neuron's activation changes with respect to the phase parameter y while keeping the input feature x constant.

(CLO-3, PLO-2, C3, 5 Marks)

b. Calculate the rate of change of the function $f(x) = (1+x)^x$

(CLO-3, PLO-2, C3, 5 Marks)

Question # 05:

During training of a simple AI model, the instantaneous training loss decreases over time t is modeled by: $L(t) = t^5 e^{-t}$ where t is the time in hours, L(t) is the loss at time t.

a. Determine the total accumulated loss $\int t^5 e^{-t} dt$

(CLO-3, PLO-2, C3, 5 Marks)

b. Calculate the loss during first two hours of training (0 to 2)

(CLO-3, PLO-2, C3, 5 Marks)

GOOD LUCK



FINAL EXAMINATION OF 1ST SEMESTER, 1st YEAR (FALL SEMESTER 2025) OF
B.S in Artificial Intelligence (A1,A2 & A3), OF 2026

FUNCTIONAL ENGLISH (THEORY)

DATED: 24-01-2026

TIME ALLOWED 02 HOURS

MAX. MARKS 25

Student Name: _____

Student Id Number: _____

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NOTE: - ATTEMPT ALL THE FOLLOWING QUESTIONS.

(CLO-01, GA-07, 10 MARKS)

Q.NO. 1 (A) Redraft the following text by filling in the blanks, where necessary with correct punctuation.

Gen Z in Pakistan is often described as confident and outspoken i) __ Their anger is frequently expressed on social media ii) __ classrooms iii) __ and public spaces. This behavior sometimes creates tension with iv) __ seniors, teachers v) __ and peers. The term respect is often discussed when talking about generational gaps. However vi) __ unchecked anger can negatively affect Gen Zs vii) __ relationships and academic growth. Can guidance and counseling help improve their behavior viii) __ Awareness programs in educational institutions are necessary ix) __ by channeling their energy into learning, innovation and self-improvement x) __ Gen Z can strengthen their relationships and academic growth. With proper guidance and counseling, they can use their potential positively for personal development and societal progress.

(B) Create sentences using the idioms below, making sure each sentence shows the meaning clearly.

i) spill the beans ii) cost an arm and a leg iii) piece of cake iv) bite the bullet v) break the ice

(CLO-02, GA-07, 07 MARKS)

Q.NO. 2 Write a letter to the Chairperson of your department requesting permission to organize an awareness workshop about new trends and technological tools in A.I. Briefly, explain the need and purpose of organizing the workshop along with a brief introduction of the trainer.

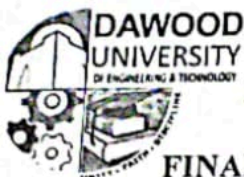
(CLO-02, GA-07, 08 MARKS)

Q.NO. 3 Using brainstorming writing strategy, draft an essay (250-300 words) on the following topic.

- Social Media and Pakistani Youth: A Tool for Awareness, Activism, and Innovation
[Underline the Thesis statement and Topic Sentences]

THE END

"AI-savvy youth are the architects of tomorrow, turning ideas into a future we've never imagined." — MHA
Best of Luck :)



DAWOOD UNIVERSITY OF ENGINEERING & TECHNOLOGY, KARACHI

FINAL EXAMINATION OF 1ST SEMESTER, 1ST YEAR (FALL SEMESTER) OF BS
ARTIFICIAL INTELLIGENCE SECTIONS (A, B & C) OF 2025F BATCH

SUBJECT: INTRODUCTION TO INFORMATION & COMMUNICATION
TECHNOLOGIES (THEORY)

DATED: 16-01-2026

TIME ALLOWED 120 MINUTES

MAX. MARKS 25

Student Name: _____

Student Id Number: _____

Instructions:

1. This paper contains 06 questions.
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NOTE: - ATTEMPT ALL THE FOLLOWING QUESTIONS.

(CLO-1, PLO-2, C-2, 03 MARKS)

Q. 1 a) Illustrate the block diagram of Data Transmission.

(CLO-1, PLO-2, C-2, 06 MARKS)

b) Describe the types of Data transmission according by method and by timing.

(CLO-1, PLO-2, C-2, 04 MARKS)

Q. 2 a) Illustrate the relationship of OSI model with TCP/IP with help of diagram.

(CLO-1, PLO-2, C-1, 04 MARKS)

b) Write the any four Commandments of Computer ethics.

(CLO-2, PLO-3, C-3, 04 MARKS)

Q. 3 a) Express the Role of IICT in Education.

(CLO-2, PLO-3, C-3, 04 MARKS)

b) Express the Role of IICT in Health care.

-----THE END-----



DAWOOD UNIVERSITY OF ENGINEERING & TECHNOLOGY, KARACHI

FINAL EXAMINATION OF 1ST SEMESTER, 1ST YEAR (FALL SEMESTER) OF
BS ARTIFICIAL INTELLIGENCE SECTIONS (A1, A2 & A3), OF 2025

SUBJECT: ISLAMIC STUDIES / ETHICS (THEORY)

DATED: 22-01-2026

TIME ALLOWED 120 MINUTES

MAX. MARKS 25

Student Name: _____

Student Id Number: _____

Instructions:

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NOTE: - ATTEMPT ALL THE FOLLOWING QUESTIONS.

ISLAMIC STUDIES AND ETHIC
QUESTION PAPER

Q1: Explain in detail which blessings of Almighty Allah were discussed in Surah-eRehman.
(CLO: 1, Level:2 ,PLO:08 Marks 08)

Q2: Explain in detail JIHAAD which reference to Surah Suff.
(CLO :1, Level:2 ,PLO:08 Marks 08)

Q3 : Outline both of the following
a) Economic system of Islam
b) Sulah Hudiabiya
(CLO:2 , Level:2, PLO:08 Marks 09)

ETHICS

Q1:Explain in detail the pillars of religion in Islam.
(CLO :1, Level:2 ,PLO:08 Marks 08)

Q2: Summarize the concept of God in buddhism
(CLO :1, Level:2 ,PLO:08 Marks 08)

Q3: Illustrate the value of Ethics and Manners in Human Society.
(CLO :2, Level:2 ,PLO:08 Marks 09)

THE END



SUBJECT: PROGRAMMING FUNDAMENTALS (THEORY)

DATED: 20-01-2026

TIME ALLOWED 03 HOURS

MAX. MARKS 50

COURSE INSTRUCTORS: SHAZMEEN BATOOL & MUHAMMAD USAMA

Student Name: _____

Student Id Number: _____

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NOTE: - ATTEMPT ALL THE FOLLOWING QUESTIONS.

Q # 1: Answer the following questions:

- a. Define which function is invoked in each case based on the given function definitions:

- i. double m (double x, double y)
- ii. double m (int x, int y)

Answer the following questions:

- A. Which of the two functions is invoked for double z = m (4, 5);
 - B. Which of the two functions is invoked for double z = m (4, 5.4);
 - C. Which of the two functions is invoked for double z = m (4.5, 5.4); [2.5 marks]
- b. Define what happens when a local variable has the same name as a global variable, and illustrate with an example. [2.5 marks]
- c. Define the difference between break and continue statements with examples. [2.5 marks]
- d. Define what happens if two overloaded functions have the same parameter types in the same order but different return types. [2.5 marks]

Q # 2: Answer the following questions:

(CLO-1, C2, 10 MARKS)

- a. Describe why we use the getline() function instead of cin to read a string from the keyboard. [2.5 marks]
- b. Describe how these definitions are invalid:
- i. float fvar;
int *iptr = &fvar;
 - ii. int *iptr = &ivar;
int ivar; [2.5 marks]

c. Describe what a global constant in C++ is and explain how it differs from a global variable. [2.5 marks]

d. Describe what is wrong with the following array definitions:

- i. `int readings[-1];`
- ii. `float measurements[4.5];` [2.5 marks]

(CLO-2, C2, 10 MARKS)

Q # 3: Answer the following questions:

a. Implement a C++ program to print an array in reverse order. [2.5 marks]

b. Implement the following program segments and write only their outputs:

- i. `string dog = "Fido";`
`cout << dog.length() << endl;`
- ii. `string s = "Hello";`
`cout << s.append(5, 's');`
- iii. `string s1 = "Programming", s2 = "Fundamentals";`
`cout << s1 + s2;` [2.5 marks]

c. Implement a C++ program that calculates the factorial of an integer number using a recursive function. The program should prompt the user to enter an integer value, and then display the factorial result. The program should include the following:

- i. A function named `factorial` that takes an integer parameter and returns the factorial of that number using recursion.
- ii. The main function should handle user input and output the result of the factorial computation. [2.5 marks]

d. Implement the following code to eliminate ambiguous invocations and explain your changes. [2.5 marks]

```
#include<iostream>
using namespace std;

int a(int x, double y)
{
    cout << x << y;
}

int a(double x, int y)
{
    cout << x << y;
}

int main()
{
    a(1,2);
}
```

(CLO-3, C3, 10 MARKS)

Q # 4: Answer the following questions:

- a. Implement a C++ program that writes the names of five cities to a file named "Hello.txt". Then, read data from the same file and print the output on the screen. Next, write the names of five countries to the same file "Hello.txt". Finally, read the data again and print all the names of cities and countries. The output should display all the city and country names. [5 marks]

- b. Implement a program that declares a structure named Student with members name, regNo, and grade. Define two variables, s1 and s2. Ask the user to input data for both variables. Then, print the data of the student whose grade is 'A'. [5 marks]

(CLO-3, C3, 10 MARKS)

Q # 5: Answer the following questions:

- a. Implement a program that takes an array of integers as input and finds the maximum and minimum numbers in the array. [5 marks]
- b. Implement a C++ program that calculates the area and circumference of a circle based on the radius entered by the user. Your program should include the following:
- Define a global constant for PI as 3.14.
 - Implement a function calculateArea that takes the radius as an argument and returns the area of the circle.
 - Implement a function calculateCircumference that takes the radius as an argument and returns the circumference of the circle.
 - In the main function, prompt the user to enter the radius of the circle.
 - Use the functions to compute the area and circumference.
 - Display the results. [5 marks]

(CLO-3, C3, 10 MARKS)

THE END